



Tutorials ▾

References ▾

Exercises ▾

Certificates ▾

Search



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[HTML](#) [CSS](#) [JAVASCRIPT](#) [SQL](#) [PYTHON](#) [JAVA](#) [PHP](#) [HOW TO](#) [W3.CSS](#) [C](#) [C++](#) [C#](#) [BOOT!](#) >

NumPy Tutorial

[NumPy HOME](#)[NumPy Intro](#)[NumPy Getting Started](#)[NumPy Creating Arrays](#)[NumPy Array Indexing](#)[NumPy Array Slicing](#)[NumPy Data Types](#)[NumPy Copy vs View](#)[NumPy Array Shape](#)[NumPy Array Reshape](#)[NumPy Array Iterating](#)[NumPy Array Join](#)[NumPy Array Split](#)[NumPy Array Search](#)[NumPy Array Sort](#)[NumPy Array Filter](#)

NumPy Random

[Random Intro](#)[Data Distribution](#)[Random Permutation](#)[Seaborn Module](#)[Normal Distribution](#)[Binomial Distribution](#)[Poisson Distribution](#)[Uniform Distribution](#)[Logistic Distribution](#)[Multinomial Distribution](#)[Exponential Distribution](#)[Chi Square Distribution](#)[Rayleigh Distribution](#)[Pareto Distribution](#)

Random Permutations

[< Previous](#)[Next >](#)

Random Permutations of Elements

A permutation refers to an arrangement of elements. e.g. [3, 2, 1] is a permutation of [1, 2, 3] and vice-versa.

The NumPy Random module provides two methods for this:

`shuffle()` and `permutation()`.

Shuffling Arrays

Shuffle means changing arrangement of elements in-place. i.e. in the array itself.

Example

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Randomly shuffle elements of following array:

```
from numpy import random
import numpy as np

arr = np.array([1, 2, 3, 4, 5])

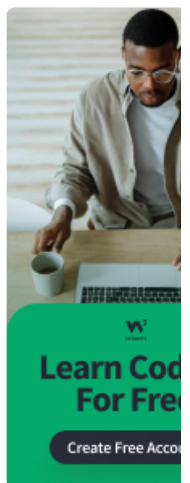
random.shuffle(arr)

print(arr)
```

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The `shuffle()` method makes changes to the original array.

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Generating Permutation of Arrays

Example

Generate a random permutation of elements of following array:

```
from numpy import random
import numpy as np

arr = np.array([1, 2, 3, 4, 5])

print(random.permutation(arr))
```

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The `permutation()` method *returns* a re-arranged array (and leaves the original array un-changed).

Exercise [?]

Which one is NOT a legal function in the Random module?

- ☐ `shuffle()`
- ☐ `mix()`
- ☐ `permutation()`

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